

Attention Proxies in SWM Kalshi News Attributions

Prediction-market exploratory analysis

June 16, 2026

Abstract

This memo asks whether external source attributes help identify news that is useful for prediction-market belief updates in the released Social World Model (SWM) Kalshi data. The main result is source-level: popularity and traffic are positive correlates of posterior helpfulness, while raw credibility labels are not. Tranco attention correlates with posterior helpfulness at Spearman 0.353, and MBFC traffic is similar (0.356). MBFC credibility is negative in the unadjusted domain panel (-0.208), a descriptive pattern that is likely confounded by source and category mix. A second diagnostic explains why source-level patterns should not be read as item-level explanations: on held-out Kalshi records, item-level prior attention and posterior helpfulness have Spearman correlation only 0.067 among records with any posterior-positive news.

1 Objective and Short Answer

The broader project asks whether prediction-market rationales that attract attention are also the rationales that add predictive information. The SWM release does not contain community rationales or comment engagement. It does contain candidate news, learned prior-attributor scores, and hindsight/posterior attribution scores. This report therefore tests the closest available source-level analogue.

- Popularity is the clearest external correlate. Tranco attention and MBFC traffic are positively associated with posterior helpfulness and prior attention.
- Credibility is not positively associated with helpfulness in the raw panel. MBFC credibility is negative for both posterior helpfulness and prior attention, but this is descriptive and likely confounded.
- The SWM prior attributor helps interpret these source patterns but is not a faithful item-level explanation. In held-out positive records, exact top-item recovery is only 16.2%.

2 What the SWM Scores Mean

The SWM paper models prediction-market prices as social beliefs and news articles as possible events that move those beliefs. For each market transition, the paper uses two attribution models. A posterior attributor Q_ϕ sees the realized next state and assigns hindsight responsibility to candidate news. A prior attributor P_η sees only the current market context and candidate news, and is trained

to approximate the posterior weights. In the released files used here, s_{ri} is the posterior/hindsight score for news item i in record r , and q_{ri} is the prior-attributor score for that same item.

$$q_{ri} = P_{\eta}(Z_r = i \mid s_r, E_r), \quad (1)$$

$$H_d = \frac{1}{N_d} \sum_{(r,i):d_{ri}=d} s_{ri}, \quad A_d = \frac{1}{N_d} \sum_{(r,i):d_{ri}=d} q_{ri}. \quad (2)$$

H_d is a domain’s average hindsight helpfulness, and A_d is the average learned prior attention assigned to that domain. The analysis uses 3,899 matched Kalshi records and 232,647 candidate news items. 1,513 records (38.8%) have at least one positive posterior attribution. We join article domains to Tranco popularity ranks, MBFC-derived traffic and credibility fields, and Iffy flags.

3 Finding 1: Popularity Is Clearer Than Credibility

External source attributes add a second distinction. Popularity measures are positive correlates of both posterior helpfulness and prior attention: Tranco attention has Spearman 0.353 with posterior helpfulness and 0.360 with prior attention. MBFC traffic is similar for posterior helpfulness (0.356). By contrast, MBFC credibility is negative in the raw domain panel (-0.208 with posterior helpfulness and -0.152 with prior attention). This does not mean credibility causes lower usefulness. The domain panel is confounded by category mix, source mix, and what kinds of events happened to move Kalshi prices in the release.

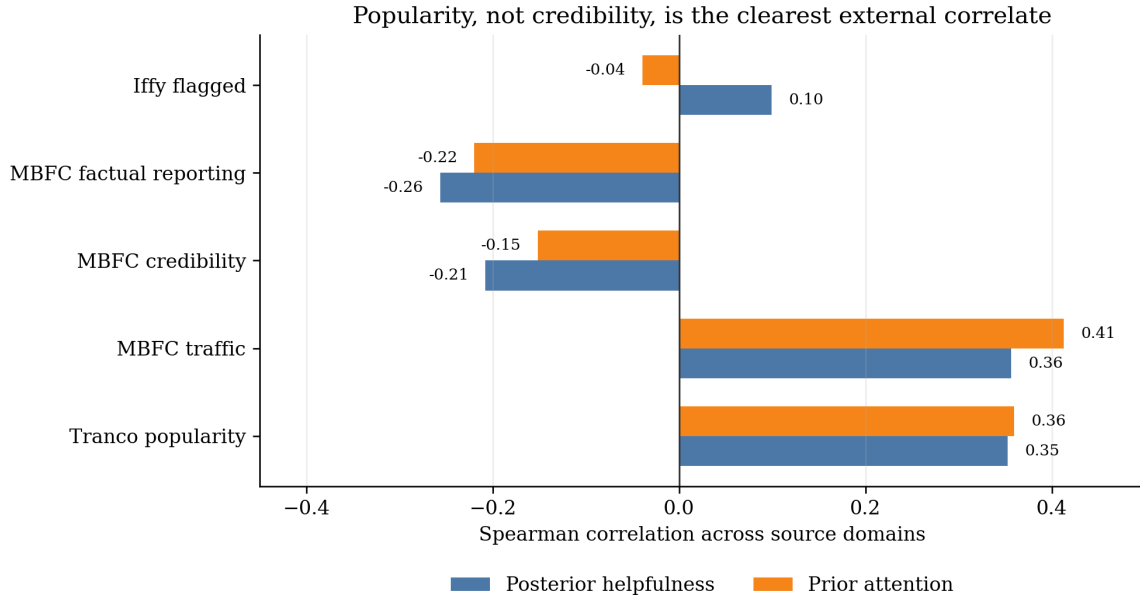


Figure 1: Bars are unadjusted Spearman correlations across source domains. Positive values mean domains higher on the attribute also have higher average posterior helpfulness or prior attention. Popularity and traffic are positive; MBFC credibility and factual-reporting labels are negative in this raw panel. These are descriptive correlations, not causal estimates.

Table 1: Unadjusted source-attribute correlations.

Predictor	Outcome	Domains	Spearman
Tranco attention	Posterior helpfulness	348	0.353
Tranco attention	Prior attention	348	0.360
MBFC credibility	Posterior helpfulness	257	-0.208
MBFC credibility	Prior attention	257	-0.152
MBFC reporting	Posterior helpfulness	257	-0.256
MBFC reporting	Prior attention	257	-0.220
MBFC traffic	Posterior helpfulness	257	0.356
MBFC traffic	Prior attention	257	0.413
Iffy flag	Posterior helpfulness	351	0.100
Iffy flag	Prior attention	351	-0.039

Spearman correlations are computed across source domains with at least 20 candidate articles. These are descriptive correlations, not causal effects.

4 Finding 2: Source-Level Attention and Helpfulness Move Together

The external-source result is consistent with the model’s own source-level behavior. Among source domains with at least 20 candidate articles, mean prior attention A_d and mean posterior helpfulness H_d have Spearman correlation 0.532 (Pearson 0.438). In other words, domains that receive more learned attention also tend to be domains whose articles receive more hindsight attribution. This is a source-level association, not proof that popularity or credibility caused the market move.

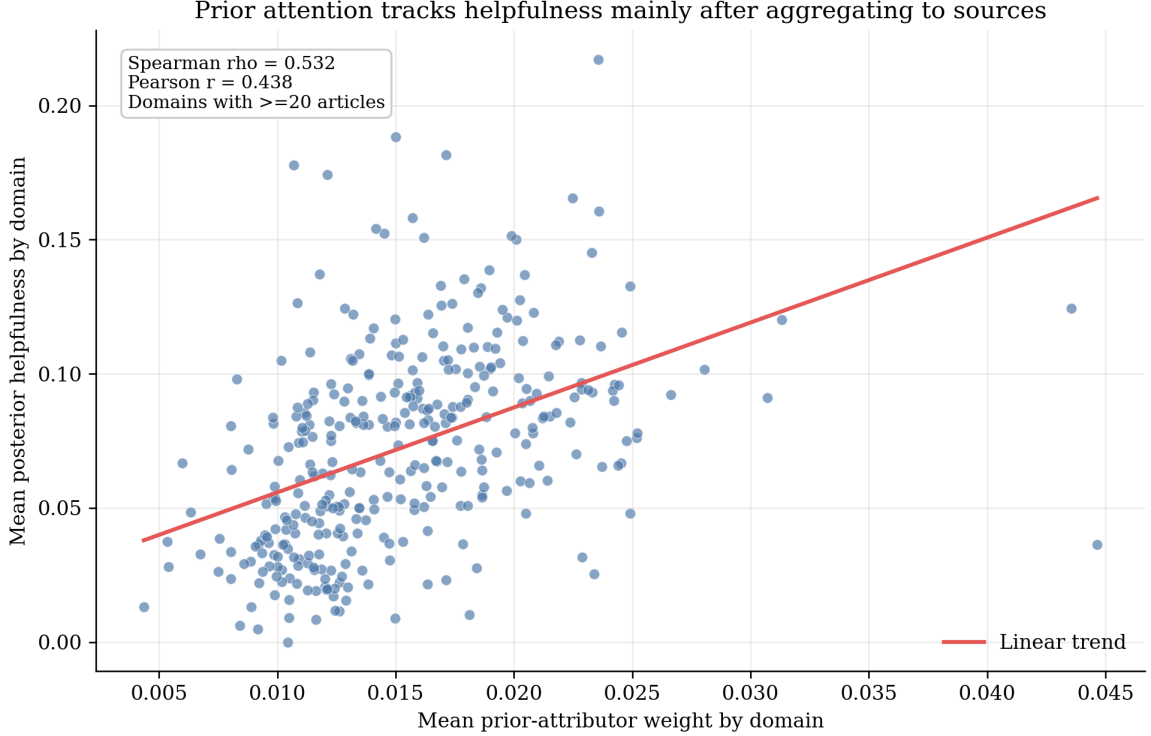


Figure 2: Each dot is a source domain with at least 20 candidate news articles. The horizontal axis is the domain’s mean prior-attributor weight A_d ; the vertical axis is its mean posterior helpfulness H_d . The positive trend shows that source-level averages line up, but it does not show causal influence or faithful item-level explanation.

5 Finding 3: Source Patterns Are Not Item-Level Explanations

The source-level correlations are easier to interpret after checking the item-level attribution behavior. Because P_η is trained to approximate posterior attributions, some relationship between q_{ri} and s_{ri} is expected. At the candidate-news level it is weak: among held-out records with any posterior-positive news, the item-level Spearman correlation between q_{ri} and s_{ri} is 0.067. The prior selects useful candidates more often than chance, but exact recovery of the hindsight top item remains low at 16.2%.

Table 2: How closely does the learned prior attributor match posterior helpfulness?

Check	Estimate	Read
Item ranking	test rho 0.067	weak item faithfulness
Within-record ranking	test mean rho 0.084	weak but positive
Mass on positive news	test mean 0.505	useful, not oracle
Top item nonzero	test 61.9%	often relevant
Exact top match	test 16.2%	low recovery
Domain averages	domain Spearman 0.532	stronger aggregated

q is the released prior-attributor score; posterior-positive means the hindsight/posterior score s is greater than zero. Test estimates use held-out Kalshi records.

6 Interpretation

The results support a careful version of the attention-is-not-information claim. Source popularity is a useful predictor of both learned attention and hindsight helpfulness in this release; credibility labels are not. But these are source-level regularities, not causal estimates and not faithful explanations of which exact article mattered. A system can learn where useful information often comes from without reliably explaining which particular item drove a belief update.

7 What This Does Not Answer

This report does not establish that a source caused a market move. It also does not test forecast-community rationales, likes, comments, author reputation, or market microstructure. Those require a Manifold- or Metaculus-style dataset with written rationales, engagement metrics, timestamps, price movement around each rationale, and final resolution. The SWM news analysis is best read as a reproducible baseline for that later rationale-level design.

A Provenance and Reproducibility

The report was generated without new LLM calls. The two analysis scripts were:

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scripts/build_source_attribute_panel.py
scripts/generate_no_training_report.py
```

The latter compiles this TeX source with the bundled Tectonic engine. The bundle includes this TeX file, a rendered PDF, figures, CSV tables, generated TeX tables, numeric macros, a Markdown report, and provenance JSON.

B Known Limits

No new LLM training or inference was run. The released Kalshi data contain candidate news, not forecast-community rationales. The posterior attribution scores are hindsight labels and should not be interpreted as formal causal proof. Actual random-news forecast harm cannot be scored without unreleased prediction outputs or a rerun of the trained world model; the available no-training proxy found a 72.9% random top-one zero-miss rate among high-attribution records.

C References

- Yu et al. (2026), *Building Social World Models with Large Language Models*: <https://arxiv.org/pdf/2606.11482>
- SWM-Bench dataset: <https://huggingface.co/datasets/ulab-ai/swm-bench>
- Social World Model code: <https://github.com/ulab-uiuc/social-world-model>
- Tranco: <https://tranco-list.eu/>
- Iffy Index: <https://iffy.news/index/>
- MBFC extension data: <https://github.com/drmikecrowe/mbfcext/>